Contact

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Top Skills

Generative AI

Linear Algebra

Statistical Learning

Languages

English (Full Professional)

Hindi (Full Professional)

Tamil (Full Professional)

Certifications

Robot Localization with Python and Particle Filters

Supervised Machine Learning: Regression

State Estimation and Localization for Self-Driving Cars

Exploratory Data Analysis for Machine Learning

Honors-Awards

First rank in National Mathematics Talent Contest

Publications

Methods and systems for ventilators

System and method for personalization and real time adaptation of spontaneous breath onset detection

Maxima-Turn-Switching Strategy of Sensor-Equipped UAVs for Target Localization in 2-D and 3-D Environments

System and method for patient-ventilator synchronization/onset detection utilizing time-frequency analysis of emg signals

Collaborative Maxima-Turn-Switching for Source Seeking in 2-

Upasana Ramakrishnan

Applied Scientist @ Amazon | GE Research | IIT Madras

Bengaluru, Karnataka, India

Summary

As an applied scientist in research, I specialize in creating innovative solutions across a range of domains, bridging the gap between cutting-edge AI technologies and practical industrial applications.

My expertise spans surrogate modeling for design optimization, industrial asset prognostics, real-time anomaly detection, causality analysis, recommendation systems and reinforcement learning for supervisory control. With a focus on leveraging machine learning models for real-world impact, I have had the opportunity to apply my skills across various industries such as wind energy, aviation and healthcare, giving me end-to-end experience from system design to hardware deployment, testing and monitoring.

Experience

Amazon

Applied Scientist

February 2025 - Present (1 year 6 months)

GE Research

5 years 8 months

Research Scientist

July 2021 - February 2025 (3 years 8 months)

Bengaluru, Karnataka, India

- Led the development of machine learning and virtual sensing based algorithms to obtain accurate real-time estimates of wind conditions experienced by turbines in a wind farm, for use in optimization of turbine operation.
- Built and tested a supervisory control platform to optimize turbine operation in real time based con wind conditions and component damage estimates, giving upto 4% gain in annual energy production.

D Environment with Two Sensor-Equipped UAVs

- Developed physics and machine learning based surrogate models for predicting wind turbine load responses and power performance to enable rapid iterations through the design space.
- Conceptualized and developed a machine learning-based inverse design framework and developed ML models for prediction of airfoil/blade performance , reducing costs of wind tunnel tests.
- Led the development of a cloud-based application for real-time decision support on component health, including design of data extraction pipeline and analytic execution. Implemented signal processing and anomaly detection algorithms on high frequency signals for identification of sensor faults and anomalies.

Edison Engineer Development Program
July 2019 - July 2021 (2 years 1 month)
Bengaluru, Karnataka, India

- Completed an intensive leadership and rotational program, supported by coursework from industry experts, to acquire advanced technical and business acumen in the development of industrial products.
- Conceptualized and implemented a Lagrangian vorticity-based method coupled with fluid-structure interaction for comprehensive analysis of aerodynamic flows over turbine blades.
- Spearheaded the development of Radial Basis Function (RBF) based global mesh morphing techniques to facilitate advanced multidisciplinary design optimization.
- Designed and executed a signal processing framework for analyzing Electromyography (EMG) signals to monitor and detect respiratory patterns in patients on mechanical ventilation.
- Developed an integrated simulation platform for wind turbine systems, enabling the testing of electrical and mechanical fault cases.

ISRO - Indian Space Research Organisation
Operational Orbital Dynamics
June 2015 - July 2015 (2 months)
Bengaluru, Karnataka, India

IEEE
Assistant Coordinator - IEEE National Level Technical Conference
October 2014 - October 2014 (1 month)
Chennai, Tamil Nadu, India

Education

Indian Institute of Technology, Madras

Master of Technology - MTech, Aerospace engineering · (2017 - 2019)

Hindustan University Chennai

Bachelor of Technology (B.Tech.), Aerospace, Aeronautical and Astronautical/ Space Engineering · (2013 - 2017)